

Salma Mohamed Sayed

Egypt, Nasr City, 7th District
Phone: +01555031062 - +01050878980
Email: salmasayed269@gmail.com
GitHub: <https://github.com/Salma269>

PROFILE

Recent graduate in **Computer Engineering and Software Systems** with a strong foundation across **AI**, software development, **embedded systems**, mobile development, and software testing. Passionate about building innovative solutions and eager to contribute to diverse technical teams. Adaptable, quick to learn, and ready to apply strong problem-solving skills to real-world challenges in any domain. Committed to continuous growth and making a tangible impact in the technology sector.

EDUCATION

Ain Shams University, Faculty of Engineering

Bachelor of Science in **Electrical Engineering, Computer Engineering, and Software Systems**

Graduated: June 2026 — GPA: 3.0

East London University, Faculty of Engineering

Bachelor of Science in **Computer Engineering and Software Systems** Program (NAF) – Dual Degree Program

Graduated: June 2026

SKILLS

- **Programming Languages:** Python, Java, TypeScript, JavaScript, C++, C, C#, SQL, Dart.
- **Frameworks & Libraries:** Flutter, React, Angular, Django, Node.js, TensorFlow, Keras, OpenCV.
- **Mobile Development:** Flutter, Dart, BLoC, Provider, Dependency Injection, GoRouter, REST APIs.
- **Web Development:** React, Angular, HTML, CSS, Bootstrap, REST APIs.
- **AI & Machine Learning:** TensorFlow, Keras, Scikit-learn, OpenCV, CNN, RNN, Optuna, SMOTE.
- **Embedded Systems:** ARM Cortex (TIVA C), Embedded C, UART, IAR Workbench, AVR.
- **Databases:** MongoDB, MySQL, Firebase, SQLite.
- **Testing & Quality:** Selenium, Postman, Flutter Testing, Mobile App Testing, JUnit.
- **DevOps & Tools:** AWS (EC2, SQS, RDS), Git, GitHub, Jira.
- **Architecture:** SOLID Principles, MVVM, MVP, Clean Architecture, Repository Pattern, Design Patterns.

EXPERIENCE

Embedded Systems Trainee – Edges For Training Academy (Aug 2025 – Sep 2025)

- Coded **AVR microcontrollers in C**, integrating LEDs, buzzers, and sensors into a responsive embedded system.
- Validated system reliability through rigorous testing on the **eta32mini AVR kit**.

React Native Trainee – ITI (Aug 2024 – Sep 2024)

- Engineered **cross-platform mobile apps** with **React Native** and **Firebase**, featuring real-time data sync and secure authentication.
- Designed responsive, high-performance UIs optimized for both **iOS and Android** platforms.

AI Trainee – Telecom Egypt (Jul 2023 – Aug 2023)

- Built and tuned **CNN/RNN** models with **TensorFlow/Keras** for object detection using **Mask R-CNN** and **OpenCV**.
- Streamlined preprocessing pipelines to boost inference speed and accuracy on video data.

GRADUATION PROJECT

FotoFlow – Document Workflow Automation Platform — Sponsored by FotopiaTech

Timeline: January 2026 – June 2026

- Built a **full-stack visual workflow automation platform** with drag-and-drop canvas, asynchronous back-end execution engine, and multi-tenant collaboration (Owner/Editor/Viewer permissions).
- Designed an extensible process catalog enabling users to select from pre-built **AI-powered nodes** with a pluggable architecture for future ML integration.
- **My Contribution – Table, Column & Cell Detection Nodes:** Engineered production-grade document parsing nodes as catalog processes, delivering end-to-end table extraction from raw invoices to structured data.
- – Achieved **95.61% accuracy** and **0.9092 IoU** for table detection using **ResNet34/EfficientNetV2-S** with

ASPP and Mish activation.

- – Built a column detection pipeline with letterboxing and fixed-pixel shrinking for preprocessing, **ResNet18-based TableNet** with **Squeeze-and-Excitation attention**, hybrid **BCE-Tversky loss** for false positive suppression, and a post-processing pipeline combining morphological erosion, connected component analysis, text-aware consolidation, and strategic boundary expansion – achieving **98.04% Precision** and **96.98% IoU**.
- – Developed a dual-strategy cell extractor combining **DBNet** with adaptive thresholding fallback, plus hierarchical grouping and gap elimination to produce perfectly aligned, gap-free tabular grids ready for OCR and export.
- The node outputs **JSON/CSV** directly into workflow execution, enabling seamless validation, export, and human-review steps.

SELECTED PROJECTS

Beacon App – Peer-to-Peer Emergency Network

Flutter, Wi-Fi Direct, SQLCipher, MVVM, STT, TTS

Jan 2026

- Engineered a decentralized Android application for infrastructure-independent peer-to-peer messaging and coordination during network outages.
- Deployed **SQLCipher** with **AES-256 encryption** for secure on-device storage of medical profiles and chat history.
- Constructed a robust **MVVM** framework integrating **STT** and **TTS** functionalities for seamless user interaction in critical situations.

Stroke Prediction – Machine Learning Pipeline

Python, Scikit-Learn, SMOTE, PCA, Optuna, SVM, K-Means

May 2025

- Developed an end-to-end pipeline to predict stroke risk using patient health records with comprehensive pre-processing and feature engineering.
- Applied **SMOTE** balancing, **PCA** dimensionality reduction, and **Optuna** hyperparameter optimization to maximize model performance.
- Achieved **97.3% accuracy** using an **RBF-SVM** model, with clustering analysis revealing distinct subgroups of high-risk patients.

Distributed Web Crawling & Indexing System

AWS (EC2, SQS, RDS), Python, Scrapy, BeautifulSoup, Whoosh, MPI

May 2025

- Designed a scalable, cloud-based distributed crawling system on **AWS** leveraging **EC2** instances and **SQS** queues for dynamic job distribution.
- Crawled and indexed over 10,000 URLs, integrating **Whoosh** full-text search engine for sub-second query performance.
- Implemented heartbeat-based node monitoring for fault tolerance and automatic recovery.

E-Commerce Platform for Cosmetics

HTML, CSS, JavaScript, Node.js, Express.js, Bootstrap, MongoDB

May 2025

- Developed a secure e-commerce platform with **Node.js** backend, **MongoDB** storage, and **JWT-based authentication**.
- Built an admin dashboard with inventory control, order management, and sales analytics.
- Designed a responsive **Bootstrap** UI with integrated user reviews, ratings, and real-time inventory updates.

Smart Home System

C, Python, Qt, Embedded C, TIVA C

Dec 2024

- Built an embedded application on **TIVA C** microcontroller using **Embedded C** and **Qt**.
- Enabled remote monitoring and control of devices like lamps, plugs, and doors with real-time GUI and **UART-based communication**.
- Included safety alert features and sensor integration.

Veri-fy Lint – Verilog Linter

Python, Verilog

Dec 2023

- Developed a Python-based parser for Veri-fy Lint.
- Addressed multiple scenarios such as arithmetic overflow, unreachable code blocks, uninitialized registers, multi-driven bus or register issues, incomplete or parallel cases, and inferred latches.

CERTIFICATIONS

- IELTS Academic – Overall Band Score: 5.5
- Embedded Systems Training – August 2025 to September 2025
- React Native Training – August 2024 to September 2024
- AI Training – July 2023 to August 2023